Adaptive Graph Convolution With Diffusion Models for Multimodal Recommendation

Jie Guo, *Senior Member, IEEE*, Ziyuan Guo, Bin Song, *Senior Member, IEEE*, and Peidong Zhang

***Abstract*—Multimodal recommendation systems (MRSs) address data sparsity and cold-start issues in collaborative filtering by incorporating multimodal item content. Recent MRSs explicitly construct item–item graphs from modality feature similarity and employ graph convolutional networks (GCNs) to propagate information for learning item representations. However, item–item graphs constructed from a single source may fail to capture complementary relationships across different information sources, while fixed propagation rules tend to blur fine-grained semantic distinctions, resulting in suboptimal recommendations. To overcome these limitations, we propose DiffGCN, a novel framework that leverages conditional diffusion models (DMs) to enable adaptive graph convolution on item–item graphs. Specifically, DiffGCN learns item-specific weights to adaptively fuse multi-source neighborhoods. On this basis, DiffGCN replaces the one-step propagation of conventional GCNs with a multi-step denoising process conditioned on neighborhood information. This iterative refinement adaptively integrates multi-source neighborhood information while preserving fine-grained semantics, yielding more expressive item representations. Extensive experiments on three real-world datasets demonstrate that DiffGCN consistently outperforms baselines while maintaining scalability, validating the effectiveness of diffusion-aware adaptive graph convolution in multimodal recommendation.**

***Index Terms*—Recommender system, multimodal recommendation, diffusion model, graph convolutional network.**

I. Introduction

RECOMMENDER systems (RSs) play a crucial role in helping users discover personalized information from the ever-growing sea of online content [1], [2], [3], [4], [5]. Collaborative filtering (CF) methods [6], [7], which learn user preferences from historical interaction such as clicks, ratings, or purchase records, have become a cornerstone of RSs. In recent years, multimodal recommendation systems (MRSs) have attracted increasing attention for their ability to incorporate multimodal content of items (e.g., images, textual descriptions) into RSs, thereby effectively alleviating the inherent data sparsity and cold start issues in CF methods [8], [9], [10], [11], [12].

Early MRSs primarily adopt feature-based fusion strategies, enhancing item representations by directly incorporating multimodal features into user–item interactions [8], [13], [14], [15], [16], [17], [18]. However, these methods typically use multimodal content to enhance the representation of individual items, while item–item relations are captured implicitly through higher-order "item-user-item" relations. More recently, structure-based MRSs [19], [20], [21] have demonstrated superior performance by explicitly modeling item–item relations. Such methods typically involve two stages: (1) constructing item–item graphs from modality feature similarity and aggregating neighborhood information via graph convolution networks (GCNs); and (2) combining the learned item representations with interactions to capture user preferences. However, structure-based MRSs continue to suffer from two inherent challenges in the process of GCN propagation over item–item graphs.

C1. Single-source neighborhoods: Existing methods typically model item–item relationships based on a single source of information, such as modality feature similarity. This leads to two key limitations. Firstly, inherent modality noise (such as irrelevant image backgrounds or redundant textual descriptions [21], [22], [23]) may introduce misleading links. Furthermore, constructing item–item graphs solely from semantic similarity overlooks additional relations, such as 'item–user–item" behavioral relations. This neglect restricts the utilization of multi-source relational information during aggregation and thereby limits the expressiveness of item representations.

C2. Fixed information propagation rules: Most existing methods adopt the LightGCN [24] propagation scheme on item–item graphs, where all nodes are updated with the same uniform aggregation rule. Such indiscriminate propagation often blurs fine-grained semantic distinctions, especially when semantically similar items share highly overlapping neighborhoods. This problem is particularly pronounced in multimodal recommendation, where fixed rules further reduce node discriminability and weaken representation quality, ultimately leading to suboptimal recommendation performance.

To address the above challenges, we propose to leverage conditional diffusion models (DMs) for unified neighborhood modeling and aggregation, enabling adaptive graph convolution. Unlike the one-step representation update in conventional

Received 29 September 2025; revised 28 April 2026; accepted 13 June 2026. Date of publication 23 June 2026; date of current version 8 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62471357 and Grant 62372357, in part by the Open Research Fund from Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ) under Grant GML-KF-24-13, in part by Scientific and Technological Innovation Teams in Shaanxi Province under Grant 2025RS-CXTD-011, and in part by ISN State Key Laboratory. Recommended for acceptance by X. He. *(Corresponding author: Bin Song.)*

The authors are with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: jguo@xidian.edu.cn; ziyuanguo@stu.xidian.edu.cn; bsong@mail.xidian.edu.cn; 25011210902@stu.xidian.edu.cn).

The source code has been released at https://github.com/ZyGuo-0708/DiffGCN.

Digital Object Identifier 10.1109/TKDE.2026.3706381

GCNs, conditional DMs adopt a multi-step generative paradigm comprising two processes [25], [26], [27]: a forward process that gradually corrupts item representations with Gaussian noise, and a reverse process that iteratively denoises the corrupted ones guided by conditioning information. Specifically, DiffGCN addresses **C1** by incorporating neighborhood representations as conditioning information in the diffusion process. Such conditioning can be flexibly constructed, allowing neighborhood information from multiple sources to be integrated within a unified framework. To address **C2**, DiffGCN departs from propagation-based aggregation by performing multi-step representation refinement via the diffusion process. The reverse process progressively refines item representations through neighborhood-guided iterative denoising, thereby preserving fine-grained semantic distinctions and producing more expressive item representations.

Based on the above insights, we propose DiffGCN, a novel multimodal recommendation framework that leverages conditional DMs for adaptive graph convolution. Firstly, DiffGCN adaptively fuses multi-source neighborhood information via learnable item-specific weights. Specifically, we consider semantic neighborhoods derived from modality feature similarity and behavioral neighborhoods constructed from high-order "item–user–item" co-occurrence relations in user interactions. Based on this, we propose a diffusion-aware representation learning module, which refines item representations through iterative denoising conditioned on neighborhood information, enabling fine-grained and adaptive item representation learning. To further improve scalability in large-scale scenarios, we develop DiffGCN-IM, an efficient variant based on the denoising diffusion implicit model (DDIM) [28]. By reducing the number of inference steps of DMs, DiffGCN-IM substantially reduces the training time of DiffGCN while maintaining competitive recommendation accuracy. The main contributions of this paper are summarized as follows:

- We propose DiffGCN, the first framework that leverages conditional DMs to achieve adaptive graph convolution, overcoming the limitations of single-source neighborhoods and fixed propagation rules in conventional GCNs.
- We propose an adaptive fusion mechanism that dynamically integrates neighborhood information from both item semantic correlations and user behavioral patterns, and iteratively refines item representations through a multi-step denoising process.
- Extensive experiments on three real-world datasets show that DiffGCN outperforms strong baselines while maintaining competitive scalability, validating the effectiveness of diffusion-aware adaptive graph convolution.

II. Related Work

A. Graph Convolutional Networks for Recommendation

Since user–item interactions can be naturally modeled as bipartite graphs, graph neural networks (GNNs) have been widely applied in recommender systems. Early studies such as NGCF [7] and GCMC [29] pioneer the use of GCNs for collaborative filtering, leveraging neighborhood aggregation to model user preferences from interaction graphs. Subsequently, LightGCN [24] simplifies the propagation mechanism, significantly improves efficiency, and becomes a representative method in recommendation. However, these approaches rely on fixed propagation over interaction graphs, which makes it difficult to capture fine-grained user preferences. To overcome this limitation, several adaptive GCN methods have emerged. For instance, AGCN [30] jointly performs item recommendation and attribute inference, adaptively refining graph embeddings with inferred attribute values. LR-GCN [31] assigns adaptive weights to different layers of GCN propagation to better capture user preferences, and CAGCN [32] dynamically adjusts neighbor weights based on the common interaction ratio. In multimodal recommendation, GRCN [15] prunes noisy edges in user–item graphs based on multimodal content to improve robustness, while LATTICE [19] adaptively updates item–item graph structures using modality features but incurs high computational and memory costs.

Although these methods demonstrate the effectiveness of adaptive graph modeling, they still rely on propagation-based aggregation and typically operate on single-source neighborhood information, which may limit their ability to capture fine-grained semantic distinctions. In contrast, DiffGCN first constructs neighborhood representations through multi-source fusion, and then refines item representations via a neighborhood-conditioned multi-step denoising process. This iterative refinement mechanism enables more effective exploitation of neighborhood information and yields more expressive representations for recommendation.

B. Multimodal Recommendation

In recent years, MRSs have demonstrated remarkable effectiveness in addressing data sparsity and cold-start issues by incorporating user–item interaction graphs with rich side information such as visual and textual modalities. Early methods like VBPR [8] adopts feature-level fusion by concatenating modality features with ID embeddings, while ACF [13] introduces attention mechanisms to capture component-level user preferences. However, these approaches often neglect high-order relations and fail to model complex user behaviors. To overcome these limitations, GNN-based architectures have been extended to MRSs by constructing modality-specific user–item or item–item graphs, enabling high-order interaction modeling and multimodal feature integration [14], [15], [16]. For example, MMGCN [14] and GRCN [15] build separate user–item bipartite graphs for each modality and perform modality-aware message propagation. DualGNN [16] further incorporates user co-occurrence graphs and modality-level attention to capture personalized user preferences across different modalities. In addition, self-supervised learning (SSL) techniques have been adopted to further enhance representation alignment and robustness [17], [18]. To further capture implicit item representations, structure-based MRSs have been proposed to explicitly model the structure between items [11], [19], [20], [21], [22], [33]. For instance, LATTICE [19] adaptively constructs item–item graphs for each modality and performs GCN-based propagation to aggregate semantic information, which is then fused into the user–item interaction graph. FREEDOM [20] further freezes the item–item graphs and adopts degree-sensitive pruning to denoise the user–item graph. More recently, MGCN [21] leverages attention mechanisms to distinguish the contributions of different modalities, leading to improved recommendation performance.

Nevertheless, most structure-based MRSs still inherit the limitations of classical GCN-based aggregation (e.g., LightGCN [24]), which relies on fixed propagation over item–item graphs and may limit the modeling of fine-grained semantics. To address this issue, we propose DiffGCN, a novel framework that leverages a learnable diffusion process to progressively refine item representations through multi-step denoising. Unlike explicit information propagation on pre-constructed item–item graphs, DiffGCN adaptively fuses multi-source neighborhood information, and then performs iterative refinement through a diffusion-aware paradigm, thereby yielding more expressive and discriminative item representations.

C. Diffusion Models for Recommendation

Diffusion models, as a novel generative paradigm, have demonstrated remarkable capabilities in modeling complex data distributions and have garnered widespread attention across diverse domains [34], [35]. Their iterative denoising mechanism enables robust distribution estimation and stable training, offering new modeling perspectives for recommender systems, which have long faced challenges such as data sparsity and complex user–item dependencies. Several pioneering studies have introduced DMs into recommendation. CODIGEM [36] and DiffRec [37] leverage the diffusion process to model discrete user–item interactions, capturing latent user preferences. Subsequently, DMs have been extended to more scenario-specific tasks, such as knowledge graph recommendation [38], social recommendation [39], and sequential recommendation [40], [41], demonstrating their adaptability and generalization across various recommendation paradigms. In the context of MRSs, the flexibility of DMs is further highlighted. MCDRec [42] directly incorporates multimodal features into item representations and denoises the user–item graph through diffusion-aware knowledge; DiffMM [43] performs diffusion on the user–item graph, injecting multimodal information to generate modality-aware user-item graphs and enhancing recommendation via self-supervised learning.

III. Preliminaries

To begin with, we briefly present the notations and problem definition, and then introduce the conditional DM based on the denoising diffusion probabilistic model (DDPM) framework [25], [26], [27].

A. Notations and Problem Definition

Notations: Let $\mathcal{U}$ and $\mathcal{I}$ denote the sets of users and items, respectively. The user-item interaction matrix is defined as $\boldsymbol{A} \in \mathbb{R}^{|\mathcal{U}| \times |\mathcal{I}|}$, where $\boldsymbol{A}_{ui} = 1$ indicates that user u has interacted with item i, and 0 otherwise. Based on this, we construct a user-item bipartite graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{\mathcal{U} \cup \mathcal{I}\}$ is the node set and $\mathcal{E} = \{(u,i)|\boldsymbol{A}_{ui} = 1\}$ is the edge set. Each item $i \in \mathcal{I}$ is associated with multimodal features $\{\boldsymbol{g}_i^m | m \in \mathcal{M}\}$, where $\mathcal{M} = \{v_m, t_m\}$ denotes the set of considered modalities (i.e., visual and textual). Note that our framework can naturally accommodate more modalities beyond the two used in this paper.

Problem Definition: Multimodal recommendation aims to match users with items they are likely to prefer, by leveraging user-item graph $\mathcal{G}$ and multimodal content $\{\boldsymbol{g}_i^m | m \in \mathcal{M}\}$. This can be naturally formulated as a two-stage solution:

- *Multimodal Representation Learning:* Getting item representations through information aggregation;
- *Preference Prediction:* Predicting user preferences based on the learned representations using a base recommender.

Formally, the objective of this paper is to learn a preference prediction function $f(\cdot)$ that estimates the preference score $\hat{y}_{ui}$ between user u and item i:

$$\hat{y}_{ui} = f(u,i) = \phi\left(\mathcal{G}, \{\mathcal{D}^m(\boldsymbol{g}_i^m)\}_{m \in \mathcal{M}}\right), \tag{1}$$

where $\mathcal{D}^m(\cdot)$ denotes the neighborhood aggregation function for modality m that generates expressive item representations, and $\phi(\cdot)$ is a base recommender (e.g., LightGCN [24]) used to predict the final preference score.

B. Conditional Diffusion Model

Given the powerful generative capabilities of DMs, we implement $\mathcal{D}^m(\cdot)$ as a diffusion process in this paper, aiming to enhance multimodal item representations via neighborhood guidance. In the following, we introduce the forward and reverse processes of the DM and the optimization strategy.

1) Forward Process and Reverse Process: During the forward process, Gaussian noise is progressively injected into the original data $\boldsymbol{x}_0$ over T discrete time steps: for $t = 1, \ldots, T$,

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = \mathcal{N}(\boldsymbol{x}_t; \sqrt{\alpha_t}\boldsymbol{x}_{t-1}, (1-\alpha_t)\boldsymbol{I}), \tag{2}$$

where $\mathcal{N}(\cdot)$ denotes the Gaussian distribution, and α_t controls the noise scale at the t-th time step. Let $\bar{\alpha}_t = \prod_{t'=1}^{t} \alpha_{t'}$, the corrupted data at the t-th time step can be directly given by:

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}(\boldsymbol{x}_t; \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0, (1-\bar{\alpha}_t)\boldsymbol{I}). \tag{3}$$

The reverse process of the conditional DM gradually reconstructs the original data $\boldsymbol{x}_0$ using a neural network parameterized by θ:

$$p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{c}) = \mathcal{N}(\boldsymbol{x}_{t-1}; \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t), \tilde{\sigma}_t^2 \boldsymbol{I}), \tag{4}$$

where $\boldsymbol{c}$ denotes the conditioning information of the reverse process, and

$$\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\boldsymbol{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t),$$
$$\tilde{\sigma}_t^2 = \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}, \tag{5}$$

where $\hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t)$ denotes the predicted $\boldsymbol{x}_0$ at time step t.

2) Optimization: With the aim of generating the original data $\boldsymbol{x}_0$, the conditional DM is optimized by maximizing the evidence lower bound of $\log p(\boldsymbol{x}_0)$. Following [27], the optimization

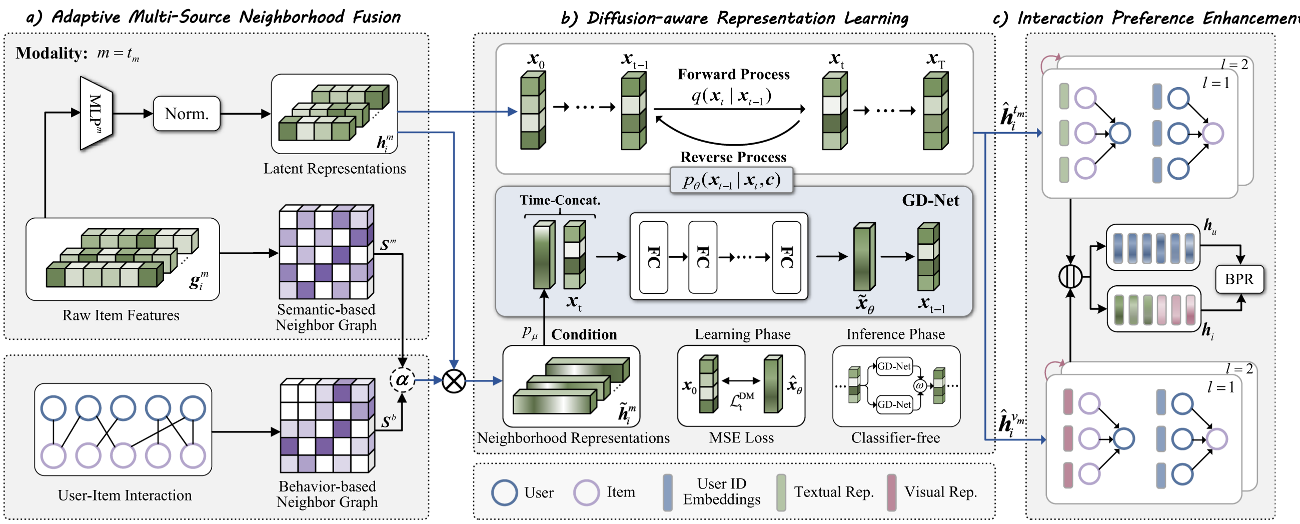


Fig. 1. The overall framework of DiffGCN. Neighborhood representations are modeled by adaptively fusing semantic and behavioral neighborhoods with learnable weights $\boldsymbol{\alpha}$, and the diffusion-aware representation learning module refines item representations through a multi-step denoising process. Then, the refined item representations are further propagated over the user–item interaction graph to enhance preference modeling. Note that (a)–(b) are illustrated in the textual modality ($m = t_m$), while the visual modality ($m = v_m$) follows the same process and is omitted here for clarity.

reduces to a weighted mean squared error (MSE) between the original data $\boldsymbol{x}_0$ and the prediction $\hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t)$ at each time step t:

$$\mathcal{L}_t^{\mathrm{DM}} = \mathbb{E}_{q(\boldsymbol{x}_t|\boldsymbol{x}_0)}\left[\hat{\alpha}_t \|\boldsymbol{x}_0 - \hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t)\|_2^2\right], \tag{6}$$

with

$$\hat{\alpha}_t = \begin{cases} 1, & t = 1, \\ \frac{1}{2}\left(\frac{\bar{\alpha}_{t-1}}{1-\bar{\alpha}_{t-1}} - \frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}\right), & t \geq 2. \end{cases} \tag{7}$$

In practice, the denoising function $\hat{\boldsymbol{x}}_\theta(\cdot)$ can be parameterized by various neural architectures, such as Multi-Layer Perceptron (MLP), U-Net [26], or Transformer [35].

IV. Proposed DiffGCN

In this section, we elaborate on our proposed DiffGCN model, which integrates neighborhood information into the denoising process of the item representations through a conditional DM, thereby achieving adaptive graph convolution and ultimately enhancing recommendation accuracy. As illustrated in Fig. 1, the DiffGCN model consists of three key components: a) Adaptive multi-source neighborhood fusion, which constructs neighborhood representations by jointly leveraging modality feature similarity and high-order "item–user–item" relations from interactions; b) diffusion-aware representation learning, which refines item representations through a conditional DM enhanced by classifier-free guidance; and c) Interaction preference enhancement, which learns user and item representations through information propagation on the interaction graph to facilitate accurate recommendation. Note that a) and b) together constitute the modality-specific neighborhood aggregation function $\mathcal{D}^m(\cdot)$ described in (1), while c) serves as the base recommender $\phi(\cdot)$.

A. Adaptive Multi-Source Neighborhood Fusion

In this section, we integrate neighborhood information from multiple sources by constructing source-specific neighbor graphs and adaptively combining them into unified neighborhood representations. In this work, we consider semantic and behavior-based information as representative examples.

1) Semantic-Based Neighbor Graph: To effectively capture the modality-specific semantic relations between items, we first calculate the cosine similarity score between each item pair for modality $m \in \{v_m, t_m\}$:

$$\tilde{S}_{ij}^m = \frac{(\boldsymbol{g}_i^m)^{\mathrm{T}} \boldsymbol{g}_j^m}{\|\boldsymbol{g}_i^m\| \|\boldsymbol{g}_j^m\|}, \tag{8}$$

where $\boldsymbol{g}_i^m$ and $\boldsymbol{g}_j^m$ are the raw features of items i and j in modality m, respectively. To promote sparsity and retain only the most semantically relevant neighbors, the kNN sparsification [44] is employed to obtain a sparse similarity matrix $\hat{\boldsymbol{S}}^m$, with each element given as:

$$\hat{S}_{ij}^m = \begin{cases} \tilde{S}_{ij}^m, & \tilde{S}_{ij}^m \in \text{top-}k(\tilde{\boldsymbol{S}}_j^m) \ \&\ i \neq j, \\ 0, & \text{otherwise}, \end{cases} \tag{9}$$

where $\tilde{\boldsymbol{S}}_j^m = [\tilde{S}_{1,j}^m, \ldots, \tilde{S}_{|\mathcal{I}|,j}^m]^{\mathrm{T}}$. Finally, we perform column-wise normalization on $\hat{\boldsymbol{S}}^m$ to derive the adjacency matrix $\boldsymbol{S}^m$ of the semantic neighbor graph, where each edge weight S_{ij}^m reflects the relative semantic importance of item i to item j under modality m:

$$S_{ij}^m = \frac{\hat{S}_{ij}^m}{\sum_{i=1}^{|\mathcal{I}|} \hat{S}_{ij}^m}. \tag{10}$$

2) Behavior-Based Neighbor Graph: To capture item relations driven by user preferences, we construct a behavior-based neighbor graph from the user-item interaction matrix. Specifically, the behavioral affinity between an item pair i and j is measured by their co-occurrence frequency in user-item interactions, denoted as $\tilde{S}^b_{ij}$:

$$\tilde{S}^b_{ij} = |\boldsymbol{U}_i \cap \boldsymbol{U}_j|, \tag{11}$$

where $\boldsymbol{U}_i$ and $\boldsymbol{U}_j$ are the sets of users who have interacted with items i and j, respectively. To mitigate the impact of noise in the interaction data, we prune low-valued elements while preserving the top-k strong connections for each item. On this basis, the sparse co-occurrence matrix $\hat{\boldsymbol{S}}^b$ can be obtained, with each element given as:

$$\hat{S}^b_{ij} = \begin{cases} \tilde{S}^b_{ij}, & \tilde{S}^b_{ij} \in \text{top-}k(\tilde{\boldsymbol{S}}^b_j) \;\&\; i \neq j \;\&\; \tilde{S}^b_{ij} > \varepsilon, \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

where ε denotes the pruning threshold. The adjacency matrix $\boldsymbol{S}^b$ of the final behavior-based neighbor graph is obtained via column-wise normalization, with each element given as:

$$S^b_{ij} = \frac{\hat{S}^b_{ij}}{\sum_{i=1}^{|\mathcal{I}|} \hat{S}^b_{ij}}. \tag{13}$$

3) Dynamic Neighborhood Representation: Building upon the constructed neighbor graphs, we define the adjacency matrix $\boldsymbol{C}^m$ of the final modality-specific neighbor graph as an item-specific weighted combination of $\boldsymbol{S}^m$ and $\boldsymbol{S}^b$:

$$\boldsymbol{C}^m = \text{diag}(\boldsymbol{\alpha}) \cdot \boldsymbol{S}^m + (\boldsymbol{I} - \text{diag}(\boldsymbol{\alpha})) \cdot \boldsymbol{S}^b, \tag{14}$$

where $\boldsymbol{\alpha} \in \mathbb{R}^{|\mathcal{I}|}$ is a learnable vector that assigns each item a personalized trade-off between semantic similarity and behavioral co-occurrence. This fusion strategy enables the model to adaptively adjust the reliance on different graph structures based on the item's characteristics during training.

To ensure dimensional consistency with user ID embeddings, the raw modality features $\{\boldsymbol{g}^m_i\}$ are projected into a shared latent space using a modality-specific MLP, followed by a normalization operation:

$$\boldsymbol{h}^m_i = \text{Norm}\left(\text{MLP}^m\left(\boldsymbol{g}^m_i\right)\right), \tag{15}$$

where $\boldsymbol{h}^m_i \in \mathbb{R}^{d\times 1}$ denotes the latent representation of item i in modality m, d is the embedding size, and $\text{Norm}(\cdot)$ represents the ℓ_2 normalization. Subsequently, the modality-specific neighborhood representation $\tilde{\boldsymbol{h}}^m_i$ for each item i can be obtained by aggregating its neighbors in the final neighbor graph $\boldsymbol{C}^m$:

$$\tilde{\boldsymbol{h}}^m_i = \left[\boldsymbol{h}^m_1, \boldsymbol{h}^m_2, \ldots, \boldsymbol{h}^m_{|\mathcal{I}|}\right] \cdot \boldsymbol{C}^m_i, \tag{16}$$

where $\boldsymbol{C}^m_i$ denotes the i-th column of $\boldsymbol{C}^m$.

B. Diffusion-Aware Representation Learning

In this section, we detail the modality-specific representation refinement process, implemented via a conditional DM introduced in Section III-B. In this framework, neighborhood representations serve as conditioning information that guide the reverse denoising process. During learning, a modality-specific Guided Denoising Network (GD-Net) is optimized using the MSE loss. During inference, the trained GD-Net refines item representations through iterative denoising, while a classifier-free guidance scheme is applied to improve generation quality.

Algorithm 1: Inference Phase of GD-Net.

Input: Input latent representation $\boldsymbol{h}^m_i$, neighborhood condition $\tilde{\boldsymbol{h}}^m_i$, and optimized parameters θ.
Output: Diffusion-aware representation $\hat{\boldsymbol{h}}^m_i$.
1: Initialize $\boldsymbol{x}_T \leftarrow \boldsymbol{h}^m_i$
2: **for** $t = T$ **to** 1 **do**
3: Get the predictions $\hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \tilde{\boldsymbol{h}}^m_i, t)$ and $\hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{0}, t)$
4: Compute $\tilde{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t)$ via (19)
5: Compute $\boldsymbol{x}_{t-1}$ using (20)
6: **end for**
7: **return** $\hat{\boldsymbol{h}}^m_i \leftarrow \boldsymbol{x}_0$

1) Learning Phase: Given a latent representation vector $\boldsymbol{h}^m_i$ of item i in modality m, the forward process is initiated by setting $\boldsymbol{x}_0 = \boldsymbol{h}^m_i$. The forward process then proceeds through T steps as $\boldsymbol{x}_0 \rightarrow \boldsymbol{x}_1 \rightarrow \cdots \rightarrow \boldsymbol{x}_T$, following (2). The corrupted data obtained at each step can be directly obtained by (3). Specifically, a linear noise schedule for $1 - \bar{\alpha}_t$ over $t \in [1, T]$ is employed, defined as:

$$1 - \bar{\alpha}_t = s \cdot \left[\alpha_{\min} + \frac{t-1}{T-1}(\alpha_{\max} - \alpha_{\min})\right], \tag{17}$$

where $s \in [0, 1]$ is a hyperparameter controlling the overall noise scale, and $\alpha_{\min}$ and $\alpha_{\max}$ denote the lower and upper bounds of the added noise, respectively. Then, in the reverse process described in (4), we leverage the neighborhood representation $\tilde{\boldsymbol{h}}^m_i$ as conditioning information $\boldsymbol{c} = \tilde{\boldsymbol{h}}^m_i$ to guide the reconstruction of the item representation.

Accordingly, for each modality $m \in \{v_m, t_m\}$, the GD-Net is optimized with the following objective:

$$\mathcal{L}^m = \frac{1}{B}\sum_{i=1}^{B} \mathcal{L}^{\text{DM}}_{t_i} \tag{18}$$

where B is the batch size, $\mathcal{L}^{\text{DM}}_{t_i}$ denotes the MSE loss at time step t for item i, as defined in (6)-(7). The GD-Net $\hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t)$ is implemented by an MLP, where the input is formed by concatenating the time embedding $\boldsymbol{e}_i$, the corrupted data $\boldsymbol{x}_t$, and the conditioning information $\boldsymbol{c} = \tilde{\boldsymbol{h}}^m_i$. Here, $\boldsymbol{e}_i$ is generated from the scalar diffusion step t by sinusoidal embedding technique.

In practical implementation, the diffusion step t is independently sampled from a uniform distribution. In addition, following the classifier-free guidance scheme [45], the conditioning information $\boldsymbol{c}$ is randomly replaced with an empty token $\boldsymbol{0}$ with probability p_μ, enabling the joint training of both conditional and unconditional DMs. Algorithm 2 summarizes the training procedure of the modality-specific semantic aggregator $\mathcal{D}^m(\cdot)$ (see lines 3-12), which is jointly trained with the base recommender $\phi(\cdot)$ using the overall objective in (26).

2) Inference Phase: In the inference phase, the trained GD-Net is utilized to generate diffusion-aware representation for each item. To enhance representation discriminability, a classifier-free guidance scheme [45] is introduced to control the influence of item neighborhood information used as conditional information. Specifically, the prediction $\hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t)$ is modified with a linear combination:

$$\tilde{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t) = (1+\omega)\hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t) - \omega\hat{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{0}, t), \tag{19}$$

where ω is a hyperparameter that controls the strength of guidance, balancing the personalized characteristics of the item with the information from its neighborhood. A higher value of ω enhances the guidance of neighborhood information, but it may lead to overfitting.

Following [26], we utilize $\boldsymbol{x}_{t-1} = \mu_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t)$ for deterministic inference. Thus, based on (4) and (5), the one-step reverse process is given by:

$$\boldsymbol{x}_{t-1} = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\boldsymbol{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t}\tilde{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t). \tag{20}$$

To generate the diffusion-aware representation for item i, we initialize the reverse process with $\boldsymbol{x}_T = \boldsymbol{h}_i^m$ and apply the reverse denoising steps $\boldsymbol{x}_T \to \boldsymbol{x}_{T-1} \to \cdots \to \boldsymbol{x}_0$ for T steps using (20), ultimately obtaining the diffusion-aware representation $\hat{\boldsymbol{h}}_i^m$. See Algorithm 1 for more details.

C. Interaction Preference Enhancement

In this section, we adopt LightGCN [24] as the base recommender $\phi(\cdot)$ to propagate representations over the user-item interaction graph in a modality-aware manner. Specifically, we define the ID embeddings of user u in modality m as $\hat{\boldsymbol{h}}_u^m \in \mathbb{R}^d$ with random initialization. For each modality $m \in \mathcal{M}$, taking the initialized user embedding $\hat{\boldsymbol{h}}_u^m$ and the diffusion-aware item representation $\hat{\boldsymbol{h}}_i^m$ (augmented with a residual connection $\boldsymbol{h}_i^m$) as the input of the user-item graph, i.e.,

$$(\bar{\boldsymbol{h}}_u^m)^0 = \hat{\boldsymbol{h}}_u^m, \;\; (\bar{\boldsymbol{h}}_i^m)^0 = \hat{\boldsymbol{h}}_i^m + \boldsymbol{h}_i^m. \tag{21}$$

Then, the representations of l-th layer can be expressed as: For $l = 1, \ldots, L$,

$$\begin{aligned}(\bar{\boldsymbol{h}}_u^m)^l &= \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u|}\sqrt{|\mathcal{N}_i|}}(\bar{\boldsymbol{h}}_i^m)^{l-1},\\ (\bar{\boldsymbol{h}}_i^m)^l &= \sum_{u \in \mathcal{N}_i} \frac{1}{\sqrt{|\mathcal{N}_u|}\sqrt{|\mathcal{N}_i|}}(\bar{\boldsymbol{h}}_u^m)^{l-1},\end{aligned} \tag{22}$$

where $\mathcal{N}_i$ and $\mathcal{N}_u$ denote the sets of users and items that interacted with item i and user u, respectively. After L layers of propagation, we aggregate representations across all layers using a simple summation:

$$\bar{\boldsymbol{h}}_u^m = \sum_{l=0}^{L}(\bar{\boldsymbol{h}}_u^m)^l, \;\; \bar{\boldsymbol{h}}_i^m = \sum_{l=0}^{L}(\bar{\boldsymbol{h}}_i^m)^l. \tag{23}$$

The final representations of users and items are then constructed by concatenating the outputs from all considered modalities $m \in \{v_m, t_m\}$:

$$\boldsymbol{h}_u = \bar{\boldsymbol{h}}_u^{v_m} || \bar{\boldsymbol{h}}_u^{t_m}, \;\; \boldsymbol{h}_i = \bar{\boldsymbol{h}}_i^{v_m} || \bar{\boldsymbol{h}}_i^{t_m}, \tag{24}$$

where $||$ denotes vector concatenation.

Algorithm 2: Training Procedure of DiffGCN. (Single Batch)

Input: For each modality $m \in \mathcal{M}$: neighbor graphs $\boldsymbol{S}^m$ and $\boldsymbol{S}^b$; a batch of raw features $\{\boldsymbol{g}_i^m | i = 1, \ldots, B\}$; sampled triplets (u, i, j) for BPR; model parameters θ'.
Output: Optimized parameters θ'.
`// Neighborhood aggregation function` $\mathcal{D}^m(\cdot)$
for each modality $m \in \mathcal{M}$ **do**
  Get $\{\boldsymbol{h}_i^m, \tilde{\boldsymbol{h}}_i^m | i = 1, \ldots, B\}$ via (14)-(16)
  for $i = 1$ **to** B **do**
    Set input $\boldsymbol{x}_0 \leftarrow \boldsymbol{h}_i^m$ and condition $\boldsymbol{c} \leftarrow \tilde{\boldsymbol{h}}_i^m$
    Sample time step $t \sim \text{Uniform}(\{1, \ldots, T\})$
    Sample noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$
    Compute $\boldsymbol{x}_t$ from $\boldsymbol{x}_0$ using (3)
    With probability p_μ, set $\boldsymbol{c} \leftarrow \boldsymbol{0}$
    Compute MSE loss $\mathcal{L}_{t_i}^{\text{DM}} \leftarrow \mathcal{L}_t^{\text{DM}}$ via (6)-(7)
  end for
  Compute DM loss $\mathcal{L}^m$ for modality m via (18)
end for
Get diffusion-aware representations $\{\hat{\boldsymbol{h}}_i^m\}$ via Algorithm 1
`// Base recommender` $\phi(\cdot)$
Compute final representations $\{\boldsymbol{h}_u, \boldsymbol{h}_i\}$ via (21)-(24)
Compute BPR loss $\mathcal{L}^{\text{BPR}}$ using (25)
Compute overall objective: $\mathcal{L}^{\text{Rec}} \leftarrow \mathcal{L}^{\text{BPR}} + \lambda\sum_{m \in \mathcal{M}}\mathcal{L}^m$
Update parameters: $\theta' \leftarrow \theta' - \eta\nabla_{\theta'}\mathcal{L}^{\text{Rec}}$

D. Optimization

Following the classical algorithms [24], [46], we adopt the Bayesian Personalized Ranking (BPR) loss as follows:

$$\mathcal{L}^{\text{BPR}} = \sum_{(u,i,j) \in \mathcal{R}} \left(-\log\sigma\left(\boldsymbol{h}_u^{\mathrm{T}}\boldsymbol{h}_i - \boldsymbol{h}_u^{\mathrm{T}}\boldsymbol{h}_j\right)\right), \tag{25}$$

where $\mathcal{R}$ denotes the set of training triplets (u, i, j) such that item i has been interacted with by user u (i.e., $\boldsymbol{A}_{ui} = 1$), while item j has not (i.e., $\boldsymbol{A}_{uj} = 0$), and $\sigma(\cdot)$ denotes the sigmoid function. To jointly enhance recommendation performance and improve the quality of item representations learned from the DM, we define the overall optimization objective as:

$$\mathcal{L}^{\text{Rec}} = \mathcal{L}^{\text{BPR}} + \lambda\sum_{m \in \mathcal{M}}\mathcal{L}^m, \tag{26}$$

where $\mathcal{L}^m$ denotes the modality-specific DM loss given in (18), and λ is a hyperparameter that balances its contribution to the overall objective. The complete training procedure of DiffGCN is summarized in Algorithm 2. Note that θ' represents the parameters of DiffGCN, including those of the neighborhood aggregation function $\mathcal{D}^m(\cdot)$ and the base recommender $\phi(\cdot)$.

V. DiffGCN-IM

In this section, we introduce DiffGCN-IM, an efficient variant that accelerates diffusion inference using the DDIM framework [28]. By reducing the number of sampling steps, DiffGCN-IM achieves significant speedup with comparable recommendation performance, which is particularly beneficial in large-scale scenarios. Then, we provide a detailed complexity analysis comparing DiffGCN and DiffGCN-IM.

A. Acceleration Via DDIM

Following [28], the original reverse update in (20) is modified as follows:

$$\boldsymbol{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\tilde{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t) + \sqrt{1-\bar{\alpha}_{t-1}} \cdot \frac{\boldsymbol{x}_t - \sqrt{\bar{\alpha}_t}\tilde{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t)}{\sqrt{1-\bar{\alpha}_t}}, \tag{27}$$

where $\tilde{\boldsymbol{x}}_\theta(\boldsymbol{x}_t, \boldsymbol{c}, t)$ is the prediction of $\boldsymbol{x}_0$ at time step t. To further improve inference efficiency of DMs, DiffGCN-IM performs generation using only S reverse steps, following $\boldsymbol{x}_{\tau_S} \rightarrow \boldsymbol{x}_{\tau_{S-1}} \rightarrow \cdots \rightarrow \boldsymbol{x}_{\tau_0}$, where $S < T$. A sparse set of time steps $\{\tau_0, \tau_1, \ldots, \tau_S\}$ is selected via uniform sampling from the original T-step DDPM schedule, using the following linear spacing rule:

$$\{\tau_i\}_{i=0}^{S} = \mathrm{linspace}(T,\ 0,\ S+1), \tag{28}$$

where linspace$(a,\ b,\ n)$ denotes n linearly spaced integers between a and b.

During the inference phase of DMs, the initial state is set to $\boldsymbol{x}_{\tau_S} = \boldsymbol{h}_i^m$, and the final diffusion-aware item representation $\hat{\boldsymbol{h}}_i^m$ is obtained by executing S DDIM-based transitions as defined in (27).

B. Complexity Analysis

The overall complexity of DiffGCN and DiffGCN-IM can be decomposed into three parts:

- *Adaptive multi-source neighborhood fusion:* This stage involves sparse matrix multiplications on the neighbor graphs, leading to a time complexity of $O(|\mathcal{I}|kd)$, where $|\mathcal{I}|$ denotes the number of items, k is the number of neighbors, and d is the embedding dimension.
- *Diffusion-aware representation learning:* The main cost lies in GD-Net. Each denoising step involves a time complexity of $O(|\mathcal{I}|h^2)$, where h is the dimensionality of the hidden representation. Accordingly, DiffGCN with T denoising steps has a complexity of $O(T|\mathcal{I}|h^2)$, whereas DiffGCN-IM reduces it to $O(S|\mathcal{I}|h^2)$ with $S < T$.
- *Interaction preference enhancing:* This stage performs message passing on the user–item graph, resulting in a complexity of $O(LPd)$, where P denotes the number of interactions and L is the number of propagation layers.

As summarized in Table I, the key difference between DiffGCN and DiffGCN-IM lies in the diffusion-aware representation learning module. By reducing the number of denoising steps from T to S, DiffGCN-IM enhances scalability by alleviating computational overhead.

TABLE I
Time Complexity Comparison Between DiffGCN and DiffGCN-IM

Variants	Time Complexity
DiffGCN	$O(\lvert I\rvert kd) + O(T\lvert I\rvert h^2) + O(LPd)$
DiffGCN-IM	$O(\lvert I\rvert kd) + O(S\lvert I\rvert h^2) + O(LPd)$

TABLE II
Statistics of the Evaluation Datasets

Datasets	#Users	#Items	#Interactions	Density
Baby	19,445	7,050	160,792	0.117%
Sports	35,598	18,357	296,337	0.045%
Electronics	192,403	63,001	1689,188	0.014%

VI. Experiments

In this section, we conduct extensive experiments to answer the following seven key research questions (RQs):

- *RQ1:* How does DiffGCN perform compared with the state-of-the-art methods for recommendation?
- *RQ2:* How much efficiency does DiffGCN-IM gain compared to DiffGCN, and what is the corresponding effect on recommendation accuracy?
- *RQ3:* How efficient of DiffGCN and DiffGCN-IM in terms of computational complexity and memory cost?
- *RQ4:* How does each component in DiffGCN influence its recommendation accuracy?
- *RQ5:* How do key parameters influence the performance of DiffGCN?
- *RQ6:* How can the DiffGCN optimize the neighborhood aggregation method of traditional GCNs?
- *RQ7:* How does DiffGCN preserve fine-grained semantic distinctions in recommendation results?

A. Experimental Settings

1) Datasets: We conduct experiments on three categories from the Amazon review datasets: *Baby*, *Sports*, and *Electronics*. Each dataset contains visual and textual information for items, with review ratings serving as implicit feedback to indicate positive user-item interactions. The raw data of each dataset is preprocessed using a 5-core setting on both items and users, as illustrated in Table II. In line with the methodology in [20], we utilize pre-trained Convolutional Neural Networks (CNNs) to extract 4096-dimensional visual features and employ sentence-transformers to obtain 384-dimensional textual embeddings.

2) Baselines: To comprehensively demonstrate the effectiveness of DiffGCN, we compared it with two groups of methods: CF methods [24], [46], and multimodal recommendation methods [8], [14], [17], [18], [19], [20], [21], [43], [47].

- BPR-MF [46] optimizes personalized rankings by factorizing the interaction matrix using a probabilistic model and minimizing a ranking-based loss function with implicit feedback.
- LightGCN [24] is the most popular GCN-based CF method, which simplifies the design of GCN to make it more appropriate for the recommendation.

- VBPR [8] directly integrates the visual features and ID embeddings of each item to model user preferences.
- MMGCN [14] constructs modality-specific interaction graphs and performs message propagation on each to learn user and item representations.
- SLMRec [17] introduces a SSL framework for multimodal recommendation, capturing multimodal item patterns through a node self-discrimination task.
- BM3 [18] simplifies the SSL framework by removing the need for randomly sampled negative examples and introducing representation perturbation via dropout.
- LATTICE [19] constructs dynamic item-item graphs based on modality feature similarity and employs LightGCN to learn enriched item representations, thereby improving recommendation performance.
- FREEDOM [20] freezes the item-item graph and denoises the interaction graph using degree-sensitive pruning.
- MGCN [21] propagates modality features purified of behavioral information across the item-item graph and employs an attention mechanism to distinguish the contributions of each modality.
- LGMRec [47] learns user and item representations from modality-specific graphs and uses hypergraph learning to jointly model local and global user preferences.
- DiffMM [43] performs diffusion on the user–item graph with multimodal information injection, and further enhances recommendation through contrastive learning.

3) Performance Metrics: To assess the accuracy of the top-K recommendation results, we adopt two widely used evaluation metrics: Recall@K (R@K) and Normalized Discounted Cumulative Gain@K (NDCG@K, denoted as N@K), where $K \in \{10, 20\}$. For each user, all non-interacted items are ranked to compute these metrics.

4) Implementation Details: Following MMRec, the embedding dimension d for both users and items is set to 64. The embedding parameters are initialized using the Xavier method, and all models are optimized with the Adam optimizer using a learning rate of 0.001. We adopt either the original implementations of the compared methods or those provided in MMRec [48], utilizing their default hyperparameter settings. All methods are implemented using PyTorch 2.8.0 and Python 3.12.9. Experiments are conducted on an NVIDIA RTX 5070 Ti GPU with 16GB of memory. We apply an early stopping strategy with a patience of 30 epochs, and the maximum number of training epochs is set to 1000. The stopping criterion is based on the R@20 metric. Each experiment is repeated with 5 different random seeds, and improvements over the best baseline are verified to be statistically significant at $p < 0.05$ using a paired t-test.

We conduct a comprehensive grid search to determine the optimal hyperparameters.

- *Neighborhood aggregation module $\mathcal{D}^m(\cdot)$:* The top-k value is fixed at $k = 10$, and the pruning threshold is set to $\varepsilon = 2$. The learnable vector $\boldsymbol{\alpha}$ is initialized with all elements set to 0.5 following [22]. For the conditional DM, the time step embedding dimension is set to 10, and the number of diffusion steps is fixed at $T = 10$. The noise scale is searched within $s \in \{0.001, 0.01\}$, the lower bound of the noise scale is fixed at $\alpha_{\min} = 0.001$, and the upper bound of the noise scale is searched within $\alpha_{\max} \in \{0.01, 0.05\}$. Following [41], the probability is set to $p_\mu = 0.1$. The strength hyperparameter is tuned within $\omega \in \{0, 2, 4, 6, 8\}$. For the DiffGCN-IM, the reverse steps is selected within $S \in \{2, 5, 8\}$.
- *Base recommender $\phi(\cdot)$:* The number of GCN layers for the user-item interaction graph is set to $L = 2$. In the overall optimization objective, λ is fixed at $1e^{-8}$.

Note that the diffusion steps T and the pruning threshold ε are fixed. The guidance scale ω is the primary hyperparameter, controlling the strength of conditioning information in the diffusion process.

B. Performance Comparison (RQ1)

To evaluate the effectiveness of DiffGCN, we perform experiments on three benchmark datasets. In Table III, the best and second-best performances are highlighted in bold and underlined, respectively. Several key observations can be drawn from the results:

DiffGCN consistently outperforms all baseline methods, including the latest diffusion-based multimodal recommendation approach, DiffMM [43]. These results highlight DiffGCN's exceptional capability in capturing user preferences and obtaining more accurate recommendations. Its advantages are twofold: (1) DiffGCN leverages dual-view neighborhood graphs to adaptively integrate semantic and behavioral relations, thus addressing the limitations of single-source neighborhoods; and (2) it performs multi-step representation learning with classifier-free guidance, thus overcoming the inflexibility of fixed propagation rules in GCNs.

DiffGCN achieves consistent gains across datasets of varying scales: Structure-based MRSs, such as LATTICE [19], FREEDOM [20], and MGCN [21], explicitly construct item–item graphs to effectively extract latent item representations, yielding strong performance on the Baby and Sports datasets. Meanwhile, SSL-based MRSs, including SLMRec [17] and BM3 [18], perform best on the larger-scale Electronics dataset (see Table II), which can be attributed to their capacity to exploit the intrinsic structure and rich feature information of large-scale data for more robust representation learning. In contrast, our proposed DiffGCN consistently demonstrates significant improvements across all datasets, underscoring its adaptability to different data scales and its ability to integrate diverse modalities effectively.

DiffGCN exhibits superior capability in aggregating neighborhood information: Traditional structure-based MRSs commonly employ GCN-based methods, such as LightGCN, to perform neighborhood aggregation on item–item graphs. However, these methods rely on uniform neighborhood propagation. In contrast, DiffGCN introduces a learnable diffusion-aware refinement mechanism that adaptively integrates semantic and behavioral neighborhood information, leading to more expressive representations and improved recommendation performance. The performance gains of DiffGCN over structure-based MRSs

TABLE III
OVERALL PERFORMANCE COMPARISON BETWEEN THE BASELINES AND DIFFGCN

	Baby				Sports				Electronics			
Methods	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**
BPR-MF	0.0357	0.0575	0.0192	0.0249	0.0432	0.0653	0.0241	0.0298	0.0235	0.0367	0.0127	0.0161
LightGCN	0.0479	0.0754	0.0257	0.0328	0.0569	0.0864	0.0311	0.0387	0.0363	0.0540	0.0204	0.0250
VBPR	0.0422	0.0664	0.0223	0.0285	0.0560	0.0856	0.0307	0.0383	0.0293	0.0458	0.0159	0.0202
MMGCN	0.0394	0.0648	0.0208	0.0273	0.0409	0.0648	0.0213	0.0275	0.0219	0.0347	0.0116	0.0149
SLMRec	0.0549	0.0838	0.0295	0.0370	0.0676	0.1017	0.0374	0.0462	<u>0.0440</u>	<u>0.0655</u>	<u>0.0248</u>	<u>0.0304</u>
BM3	0.0541	0.0859	0.0294	0.0376	0.0635	0.0970	0.0348	0.0434	0.0434	0.0636	0.0244	0.0296
LATTICE	0.0551	0.0850	0.0292	0.0369	0.0620*	0.0953*	0.0335*	0.0421*	-	-	-	-
FREEDOM	0.0621	<u>0.0998</u>	0.0328	0.0425	0.0711	0.1077	0.0384	0.0478	0.0400	0.0603	0.0220	0.0273
MGCN	0.0607	0.0950	0.0327	0.0416	<u>0.0736</u>	<u>0.1105</u>	<u>0.0404</u>	<u>0.0499</u>	0.0354	0.0538	0.0196	0.0244
LGMRec	<u>0.0651</u>	0.0987	<u>0.0352</u>	<u>0.0438</u>	0.0700	0.1054	0.0380	0.0471	0.0394	0.0600	0.0218	0.0272
DiffMM	0.0605	0.0943	0.0322	0.0407	0.0705	0.1044	0.0377	0.0464	0.0393	0.0596	0.0204	0.0254
DiffGCN	**0.0697**	**0.1053**	**0.0380**	**0.0471**	**0.0773**	**0.1163**	**0.0413**	**0.0514**	**0.0471**	**0.0697**	**0.0262**	**0.0320**
Improv.	7.07%	5.51%	7.95%	7.53%	5.03%	5.25%	2.23%	3.01%	7.05%	6.41%	5.65%	5.26%

'-' indicates that the method cannot be fitted into a NVIDIA RTX5070 Ti GPU card with 16GB memory.
* denotes results are copied from FREEDOM [20].

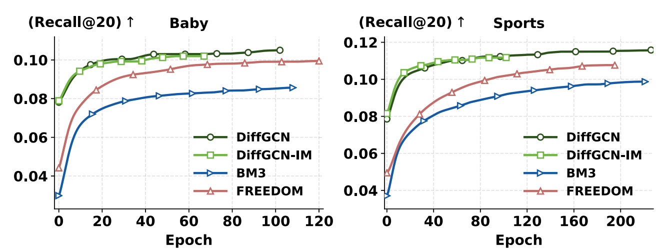


Fig. 2. Comparison of the number of training epochs required for convergence: DiffGCN, DiffGCN-IM, BM3, and FREEDOM.

TABLE IV
PERFORMANCE AND EFFICIENCY COMPARISON BETWEEN DIFFGCN AND DIFFGCN-IM

Datasets	**Methods**	**R@20↑**	**N@20↑**	**Time (s/epoch)↓**
Baby	DiffGCN	0.1053	0.0471	2.67
	DiffGCN-IM	$0.1034^{-1.8\%}$	$0.0459^{-2.5\%}$	$2.21^{-17.2\%}$
Sports	DiffGCN	0.1163	0.0514	7.37
	DiffGCN-IM	$0.1153^{-0.9\%}$	$0.0517^{+0.6\%}$	$6.68^{-9.4\%}$
Elec.	DiffGCN	0.0697	0.0320	131.19
	DiffGCN-IM	$0.0692^{-0.7\%}$	0.0320^{-}	$71.66^{-45.4\%}$

validate the effectiveness of conditional diffusion models as a powerful alternative to conventional GCNs for item neighborhood aggregation.

C. DiffGCN vs. DiffGCN-IM (RQ2)

DiffGCN-IM significantly reduces the time complexity of DiffGCN by performing fewer inference steps S of DMs within the DDIM framework. To balance effectiveness and efficiency, we empirically set $S = 5$ for the Baby dataset, $S = 8$ for the Sports dataset, and $S = 2$ for the Electronics dataset. Table IV presents a comparative analysis of recommendation performance and training time between DiffGCN and DiffGCN-IM across the three datasets, showing that DiffGCN-IM achieves substantial efficiency gains with minimal performance loss. On the Baby dataset, it incurs a moderate decline in R@20 and N@20 (1.8% and 2.5%, respectively, highlighted in green), whereas on the Sports and Electronics datasets the performance drop remains within 1%, and in some cases even slightly improves (highlighted in red). Notably, on the large-scale Electronics dataset, DiffGCN-IM achieves up to 45.4% training time reduction (highlighted in blue) with less than 1% accuracy loss, demonstrating its practical effectiveness for large-scale recommendation scenarios.

D. Efficiency Analysis (RQ3)

1) Training Efficiency: To provide a comprehensive evaluation of the training efficiency of DiffGCN, we analyze both per-epoch training cost and total training cost to convergence.

Per-epoch training cost: Table V reports the memory consumption and training time per epoch of DiffGCN (DiffGCN-IM) and representative baselines. Compared to recent MRSs, DiffGCN achieves the lowest memory consumption, benefiting from its core design of diffusion-aware representation learning. Unlike other structure-based MRSs that require explicit propagation across the entire item-item graph to aggregate item semantics, DiffGCN only performs diffusion-aware refinement at the representation level within each batch. Despite employing complex DMs, DiffGCN exhibits notably lower training time than the recently proposed DiffMM, which performs a diffusion process on the interaction graph. Furthermore, DiffGCN-IM further accelerates training by reducing the number of inference steps, achieving efficiency comparable to state-of-the-art baselines.

Total training cost: To provide a more comprehensive evaluation of training efficiency, we further report the total training time to convergence and analyze the convergence behavior of DiffGCN and its variants. Table VI presents the total training time required for convergence of DiffGCN (DiffGCN-IM) and representative baselines. Although DiffGCN incurs higher per-epoch computational cost due to the multi-step denoising process, DiffGCN-IM significantly reduces the overall training time by decreasing the number of diffusion steps while

TABLE V
COMPARISON OF DIFFGCN AND DIFFGCN-IM AGAINST STATE-OF-THE-ART BASELINES ON MODEL EFFICIENCY

Datasets	Metrics	Traditional MRSs						Diffusion-based MRSs		
		SLMRec	BM3	LATTICE	FREEDOM	MGCN	LGMRec	DiffMM	DiffGCN	DiffGCN-IM
Baby	Memory (GB)	1.91	2.18	4.26	2.20	2.31	2.05	2.68	1.88	1.88
	Time (s/epoch)	1.96	1.62	2.01	1.64	2.96	2.23	3.48	2.67	2.21
Sports	Memory (GB)	2.69	3.38	-	3.41	3.62	3.29	4.37	2.67	2.67
	Time (s/epoch)	4.50	3.69	-	3.89	4.78	5.40	11.58	7.37	6.68
Electronics	Memory (GB)	6.67	8.34	-	8.47	9.69	7.94	14.46	6.38	6.38
	Time (s/epoch)	82.86	45.02	-	44.51	69.77	104.97	287.72	131.19	71.66

'-' indicates that the method cannot be fitted into a NVIDIA RTX5070 Ti GPU card with 16GB memory.

TABLE VI
COMPARISON OF TOTAL TRAINING TIME TO CONVERGENCE (S) FOR DIFFGCN, DIFFGCN-IM, AND BASELINES

Datasets	BM3	FREEDOM	LGMRec	DiffGCN	DiffGCN-IM
Baby	164.2	192.2	96.3	278.5	152.1
Sports	782.3	731.3	375.1	1665.6	684.4

TABLE VII
COMPARISON OF INFERENCE TIME (S/EPOCH) AMONG DIFFGCN, DIFFGCN-IM, AND BASELINES

Datasets	BM3	FREEDOM	LGMRec	DiffGCN	DiffGCN-IM
Baby	1.12	1.12	1.14	1.40	1.26
Sports	2.05	2.08	2.12	3.10	2.92
Electronics	49.28	42.96	98.99	131.39	69.12

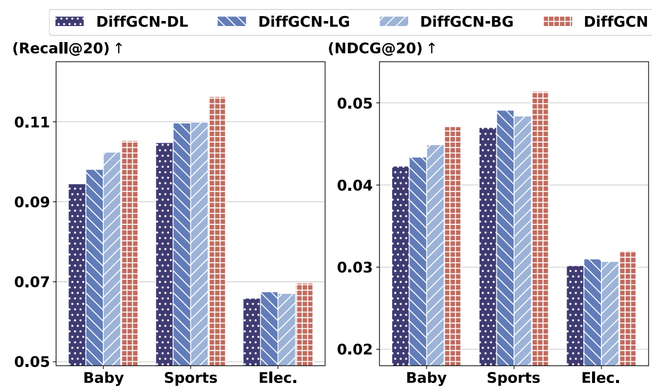


Fig. 3. Performance comparison between different variants of DiffGCN.

maintaining comparable performance. Specifically, on the Baby and Sports datasets, DiffGCN-IM reduces the total training time of DiffGCN by 45.4% and 58.9%, respectively. Notably, DiffGCN-IM also achieves lower total training time than BM3 and FREEDOM, demonstrating that the proposed variant effectively controls the optimization cost while maintaining strong performance. Fig. 2 illustrates the convergence curves of DiffGCN, DiffGCN-IM, and representative baselines. Throughout most of the training process, both DiffGCN variants consistently achieve higher Recall@20 than the baselines, indicating more effective optimization dynamics. In addition, DiffGCN converges within a comparable number of epochs to baseline models, while DiffGCN-IM further accelerates convergence and achieves similar performance. These observations are consistent with the total training time results and further validate the favorable efficiency–performance trade-off of the proposed method.

2) Inference Efficiency: We further evaluate the inference efficiency of DiffGCN and DiffGCN-IM. Table VII reports the inference time of DiffGCN and representative baselines. Due to the multi-step denoising process, DiffGCN introduces additional inference overhead. In contrast, DiffGCN-IM significantly improves efficiency by reducing the number of reverse steps, achieving inference time comparable to or better than representative baselines. Specifically, on the Baby, Sports, and Electronics datasets, DiffGCN-IM reduces the inference time of DiffGCN by 10.0%, 5.8%, and 47.4%, respectively. These results demonstrate that DiffGCN-IM effectively mitigates the computational cost of diffusion while maintaining high scalability for large-scale recommendation scenarios.

E. Ablation Study (RQ4)

1) Effect of Modules: To evaluate the contribution of individual components in DiffGCN, we perform ablation studies by removing or replacing three key modules as follows:

- *DiffGCN-DL (w/o Diffusion Learning):* This variant removes the diffusion-aware representation learning module and directly propagates latent representations on the interaction graph.
- *DiffGCN-LG (replace with LightGCN):* This variant removes the diffusion-aware representation learning module and instead uses LightGCN to propagate information over modality-specific item-item graphs for multimodal representation learning.
- *DiffGCN-BG (w/o Behavior-based Neighbor Graph):* This variant removes the behavior-based neighbor graph when constructing neighborhood representations.

As shown in Fig. 3, the **DiffGCN-DL** variant, which removes the diffusion-aware representation learning module, exhibits the most pronounced performance degradation across all datasets. This highlights the critical role of conditional diffusion models in enabling effective item neighborhood aggregation. **DiffGCN-LG**, which replaces the diffusion-aware module with LightGCN, also shows a notable performance drop. This result further confirms the superiority of conditional DMs over traditional GCNs, as DMs offer flexibility in aggregating neighborhood

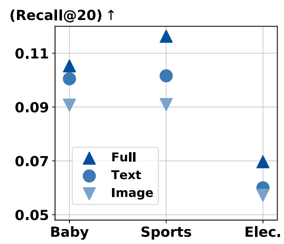


Fig. 4. Performance comparison across different modalities on three datasets.

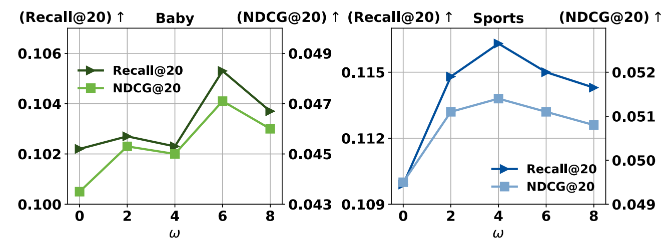


Fig. 5. Performance comparison under different ω on the Baby and Sports datasets.

TABLE VIII
PERFORMANCE COMPARISON UNDER DIFFERENT s AND α_{MAX}

		Baby		Sports		Electronics	
s	$\alpha_{\max}$	R@20	N@20	R@20	N@20	R@20	N@20
0.001	0.01	**0.1053**	**0.0471**	0.1142	0.0509	0.0687	0.0315
	0.05	0.1040	0.0463	0.1149	0.0510	0.0680	0.0312
0.01	0.01	0.1048	0.0461	**0.1163**	**0.0514**	0.0694	0.0318
	0.05	0.1017	0.0452	0.1151	0.0513	**0.0697**	**0.0320**

information and provide fine-grained representation modeling, thereby addressing the issues of single-source neighborhoods and inflexible propagation rules. The **DiffGCN-BG** variant, which removes the behavior-based neighbor graph, exhibits clear performance drops across all datasets, fully demonstrating the effectiveness of adaptively fusing multi-source neighborhoods. The decline is more pronounced on the Sports and Electronics datasets than on Baby, suggesting that rich behavioral information provides more valuable higher-order item–item relations in these domains, which are essential for learning expressive representations.

2) Effect of Modalities: To evaluate the contribution of each modality to recommendation performance, we conducted experiments under different modality input conditions. As shown in Fig. 4, the best performance is achieved when all modalities are utilized (Full), indicating that DiffGCN effectively integrates textual and visual information to capture user preferences. This result also validates the effectiveness of the proposed diffusion-aware representation learning. Furthermore, across all datasets and evaluation metrics, the text-only setting consistently outperforms the image-only counterpart, suggesting that textual features generally convey richer semantic information that is more beneficial for preference modeling.

F. Sensitivity Analysis (RQ5)

1) Effect of ω in the Inference Phase: The hyperparameter ω in (19) plays a pivotal role in the classifier-free guidance scheme, controlling the strength of conditional information during the denoising process. In DiffGCN, it determines the guidance strength of neighborhood information for item nodes, thereby balancing node-specific representations with contextual information from their neighborhoods. Careful tuning of ω is essential, as overly large values may hinder the generalization capacity of the DM and result in low-quality multimodal item representations.

As shown in Fig. 5, we present the experimental results on the Baby and Sports datasets under varying values of ω. When $\omega = 0$, both the Baby and Sports datasets exhibit substantially lower performance, highlighting the importance of the classifier-free guidance scheme in enhancing generation quality. In addition, increasing the value of ω initially improves recommendation accuracy, aligning with the intuition that stronger neighborhood information can enhance neighborhood aggregation. However, excessively high values cause performance degradation (e.g., $\omega = 8$), suggesting that overemphasizing conditioning information risks overfitting and reduces generation quality. Although the results for the Electronics dataset are omitted for brevity, it follows the same trend, with optimal performance achieved at $\omega = 6$.

2) Effect of s and $\alpha_{\max}$ in the Learning Process: Table VIII reports the impact of different s and $\alpha_{\max}$ settings on recommendation performance. The results show that the optimal noise scale is $s = 0.01$ for the Sports and Electronics datasets, whereas a smaller value of $s = 0.001$ yields the best performance on the smaller Baby dataset. Moreover, a larger noise span leads to better results on the large-scale Electronics dataset. These results suggest that larger datasets benefit from higher noise scales, as stronger perturbations are required to sufficiently explore and capture their more complex and diverse data distributions.

G. Visualization Analysis (RQ6)

To evaluate the effectiveness of DiffGCN in neighborhood aggregation, we compare multimodal text representations generated by Algorithm 1 with those produced by LightGCN (the DiffGCN-LG variant in Section VI-E) through convolution on the text-based item–item graph. We first apply t-SNE [49] to map the high-dimensional representations into a 2-dimensional space, and then employ Gaussian kernel density estimation (KDE) to visualize their spatial distribution, as shown in Fig. 6. For clearer interpretation, the KDE on the unit hypersphere S^1 is shown at the bottom of each figure. As illustrated, the representations learned by DiffGCN exhibit a more uniform and well-dispersed distribution on both the Baby and Sports datasets, suggesting that each item obtains a more distinctive and discriminative representation. In contrast, the DiffGCN-LG variant produces highly condensed clusters, reflecting the limitations of fixed propagation in GCNs, which often lead to representation degradation and over-smoothing. These results confirm the superior ability of conditional DMs to adaptively integrate neighborhood information and preserve fine-grained

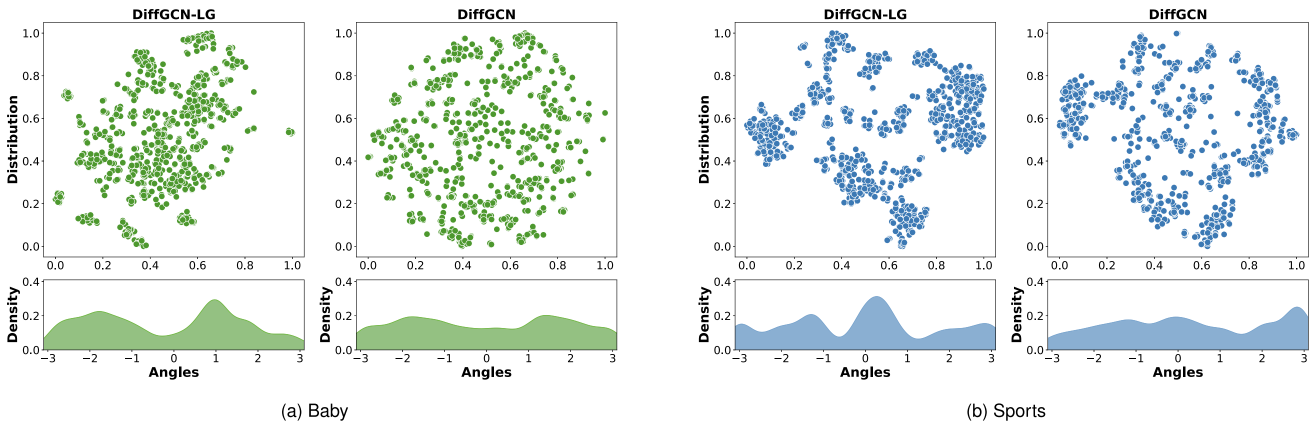


Fig. 6. Distribution of multimodal text representations for the Baby and Sports datasets. In each subfigure, the left part shows the distributions obtained by DiffGCN-LG using LightGCN propagation on the item–item graph, while the right shows the distributions generated by DiffGCN through diffusion-aware representation learning.

semantics during aggregation. Moreover, this observation aligns with prior findings [21], [50] that representation uniformity on the embedding space benefits recommendation accuracy, as it reduces redundancy and improves representation discriminability.

From a representation learning perspective, the observed difference can be attributed to the distinct update mechanisms of GCN-based propagation and diffusion-based refinement. Conventional GCN aggregation propagates and aggregates neighbor information, which may drive representations toward a low-rank subspace, potentially leading to over-smoothing. In contrast, DiffGCN performs iterative refinement through neighborhood-conditioned denoising, where each reverse step progressively adjusts representations. This correction-based update mechanism mitigates representation homogenization and helps preserve fine-grained semantic distinctions, which is consistent with the more dispersed representation distribution observed above.

H. Case Study (RQ7)

To further investigate whether DiffGCN preserves fine-grained semantic distinctions, we present a user-oriented case study on the Baby and Sports datasets. We compare DiffGCN with its variant DiffGCN-LG, where the diffusion-aware representation learning module is removed and replaced with LightGCN-based propagation over modality-specific item–item graphs.

As shown in Fig. 7, we present two representative cases, where each row presents the top-5 items ranked by the corresponding model. In the Baby case, the target item of user 12632 is item `905` (take-along toy). DiffGCN not only successfully retrieves the target item, but also produces semantically coherent recommendations (e.g., music toy, toy links, cloth book), indicating that it preserves fine-grained semantics related to infant companion toys. In contrast, DiffGCN-LG tends to drift toward coarser infant-toy categories (e.g., crib mirror, links rattle), failing to capture more specific semantics. In the Sports case, the target item of user 10709 is item `799` (*wrist-wrap glove*). DiffGCN ranks the target item highly and retrieves semantically

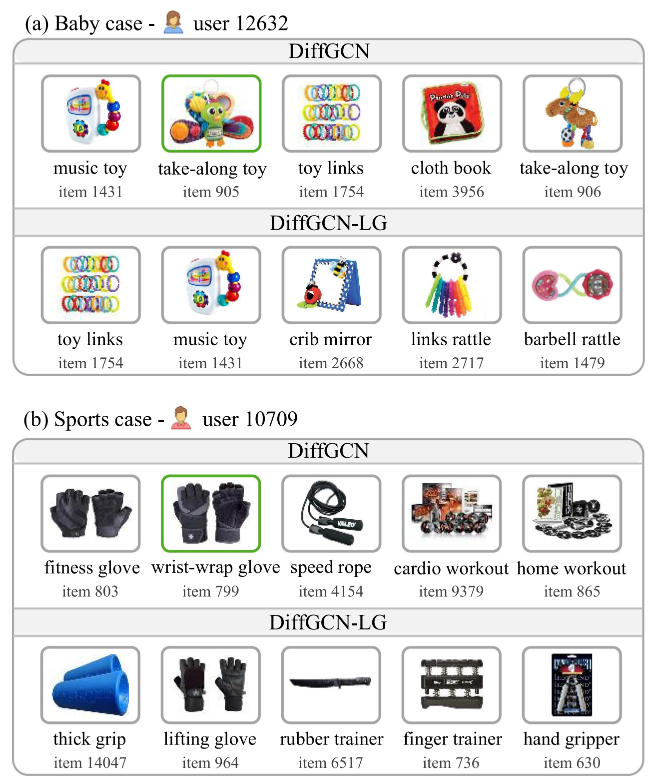


Fig. 7. Case studies on the Baby and Sports datasets. The green box denotes the ground-truth item hit in the recommendation list. Each row shows the top-5 recommended items ranked by the corresponding model.

aligned items such as fitness glove, reflecting accurate modeling of glove-specific semantics with wrist support. By contrast, DiffGCN-LG shifts toward a broader semantic region of hand- or grip-related products (e.g., finger trainer, hand gripper), which are less consistent with the user's fine-grained preference.

Overall, this case study provides direct empirical evidence that DiffGCN better preserves fine-grained semantic neighborhoods. Compared with propagation-based methods, DiffGCN reduces semantic drift and enables more precise retrieval of user-relevant items.

VII. Conclusion and Future Work

In this paper, we proposed DiffGCN, the first framework that leverages conditional diffusion models for adaptive graph convolution in multimodal recommendation. By adaptively integrating semantic and behavioral neighborhood information into the diffusion process, DiffGCN overcomes the inherent limitations of single-source neighborhood structures and fixed propagation rules in traditional GCN-based approaches. Extensive experiments on three real-world datasets demonstrate that DiffGCN consistently outperforms strong baselines while maintaining competitive scalability. Moreover, the proposed DiffGCN-IM variant further improves inference efficiency, making the framework well-suited for large-scale recommendation scenarios.

In future work, we plan to further investigate the advantages of diffusion models for graph convolution and extend their applications to broader recommendation scenarios. Specifically, the adaptive denoising paradigm of diffusion can be generalized beyond isomorphic item–item graphs to heterogeneous graphs, where relation types and graph structures are more complex. By leveraging the flexibility of diffusion-aware aggregation, we aim to provide new perspectives for advancing graph-based recommendation methods.

References

[1] Y. Shang, C. Gao, J. Chen, D. Jin, M. Wang, and Y. Li, "Learning fine-grained user interests for micro-video recommendation," in *Proc. 46th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2023, pp. 433–442.

[2] B. Kersbergen, O. Sprangers, and S. Schelter, "Serenade - Low-latency session-based recommendation in E-commerce at scale," in *Proc. 2022 Int. Conf. Manage. Data*, 2022, pp. 150–159.

[3] C. Gao, T.-H. Lin, N. Li, D. Jin, and Y. Li, "Cross-platform item recommendation for online social E-commerce," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 2, pp. 1351–1364, Feb. 2023.

[4] U. Singer et al., "Sequential modeling with multiple attributes for watchlist recommendation in E-commerce," in *Proc. 15th ACM Int. Conf. Web Search Data Mining*, 2022, pp. 937–946.

[5] J. Du, S. Zhou, J. Yu, P. Han, and S. Shang, "Cross-task multimodal reinforcement for long tail next POI recommendation," *IEEE Trans. Multimedia*, vol. 26, pp. 1996–2005, 2024.

[6] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T.-S. Chua, "Neural collaborative filtering," in *Proc. 26th Int. Conf. World Wide Web*, 2017, pp. 173–182.

[7] X. Wang, X. He, M. Wang, F. Feng, and T.-S. Chua, "Neural graph collaborative filtering," in *Proc. 42nd Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2019, pp. 165–174.

[8] R. He and J. McAuley, "VBPR: Visual Bayesian personalized ranking from implicit feedback," in *Proc. AAAI Conf. Artif. Intell.*, 2016, pp. 144–150.

[9] Y. Zhou, J. Guo, H. Sun, B. Song, and F. R. Yu, "Attention-guided multi-step fusion: A hierarchical fusion network for multimodal recommendation," in *Proc. 46th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2023, pp. 1816–1820.

[10] Y. Zhou, J. Guo, B. Song, C. Chen, J. Chang, and F. R. Yu, "Trust-aware multi-task knowledge graph for recommendation," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 8, pp. 8658–8671, Aug. 2023.

[11] J. Guo, L. Wen, Y. Zhou, B. Song, Y. Chi, and F. R. Yu, "SPACE: Self-supervised dual preference enhancing network for multimodal recommendation," *IEEE Trans. Multimedia*, vol. 26, pp. 8849–8859, 2024.

[12] Z. Wang, Y. Feng, X. Zhang, R. Yang, and B. Du, "Multi-modal correction network for recommendation," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 2, pp. 810–822, Feb. 2025.

[13] J. Chen, H. Zhang, X. He, L. Nie, W. Liu, and T.-S. Chua, "Attentive collaborative filtering: Multimedia recommendation with item- and component-level attention," in *Proc. 40th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2017, pp. 335–344.

[14] Y. Wei, X. Wang, L. Nie, X. He, R. Hong, and T.-S. Chua, "MMGCN: Multi-modal graph convolution network for personalized recommendation of micro-video," in *Proc. 27th ACM Int. Conf. Multimedia*, 2019, pp. 1437–1445.

[15] Y. Wei, X. Wang, L. Nie, X. He, and T.-S. Chua, "Graph-refined convolutional network for multimedia recommendation with implicit feedback," in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 3541–3549.

[16] Q. Wang, Y. Wei, J. Yin, J. Wu, X. Song, and L. Nie, "DualGNN: Dual graph neural network for multimedia recommendation," *IEEE Trans. Multimedia*, vol. 25, pp. 1074–1084, 2021.

[17] Z. Tao et al., "Self-supervised learning for multimedia recommendation," *IEEE Trans. Multimedia*, vol. 25, pp. 5107–5116, 2023.

[18] X. Zhou et al., "Bootstrap latent representations for multi-modal recommendation," in *Proc. ACM Web Conf. 2023*, 2023, pp. 845–854.

[19] J. Zhang, Y. Zhu, Q. Liu, S. Wu, S. Wang, and L. Wang, "Mining latent structures for multimedia recommendation," in *Proc. 29th ACM Int. Conf. Multimedia*, 2021, pp. 3872–3880.

[20] X. Zhou and Z. Shen, "A tale of two graphs: Freezing and denoising graph structures for multimodal recommendation," in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 935–943.

[21] P. Yu, Z. Tan, G. Lu, and B.-K. Bao, "Multi-view graph convolutional network for multimedia recommendation," in *Proc. 31st ACM Int. Conf. Multimedia*, 2023, pp. 6576–6585.

[22] G. Xv et al., "Improving multi-modal recommender systems by denoising and aligning multi-modal content and user feedback," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2024, pp. 3645–3656.

[23] J. Zhang, Y. Zhu, Q. Liu, M. Zhang, S. Wu, and L. Wang, "Latent structure mining with contrastive modality fusion for multimedia recommendation," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 9, pp. 9154–9167, Sep. 2023.

[24] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, and M. Wang, "LightGCN: Simplifying and powering graph convolution network for recommendation," in *Proc. 43 rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2020, pp. 639–648.

[25] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," in *Proc. 32nd Int. Conf. Mach. Learn.*, 2015, vol. 37, pp. 2256–2265.

[26] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. 34th Conf. Neural Inf. Process. Syst.*, 2020, pp. 6840–6851.

[27] C. Luo, "Understanding diffusion models: A unified perspective," 2022, *arXiv:2208.11970*.

[28] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Proc. Int. Conf. Learn. Representations*, 2021.

[29] R. Van Den Berg, T. N. Kipf, and M. Welling, "Graph convolutional matrix completion," 2017, *arXiv:1706.02263*.

[30] L. Wu, Y. Yang, K. Zhang, R. Hong, Y. Fu, and M. Wang, "Joint item recommendation and attribute inference: An adaptive graph convolutional network approach," in *Proc. 43rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2020, pp. 679–688.

[31] X. Zhou, D. Lin, Y. Liu, and C. Miao, "Layer-refined graph convolutional networks for recommendation," in *Proc. IEEE 39th Int. Conf. Data Eng.*, 2023, pp. 1247–1259.

[32] Y. Wang, Y. Zhao, Y. Zhang, and T. Derr, "Collaboration-aware graph convolutional network for recommender systems," in *Proc. ACM Web Conf. 2023*, 2023, pp. 91–101.

[33] Z. Mu, Y. Zhuang, J. Tan, J. Xiao, and S. Tang, "Learning hybrid behavior patterns for multimedia recommendation," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 376–384.

[34] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. 35th Conf. Neural Inf. Process. Syst.*, 2021, pp. 8780–8794.

[35] X. Li, J. Thickstun, I. Gulrajani, P. S. Liang, and T. B. Hashimoto, "Diffusion-LM improves controllable text generation," in *Proc. 36th Conf. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 4328–4343.

[36] J. Walker, T. Zhong, F. Zhang, Q. Gao, and F. Zhou, "Recommendation via collaborative diffusion generative model," in *Proc. Int. Conf. Knowl. Sci., Eng. Manage.*, 2022, pp. 593–605.

[37] W. Wang, Y. Xu, F. Feng, X. Lin, X. He, and T.-S. Chua, "Diffusion recommender model," in *Proc. 46th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2023, pp. 832–841.

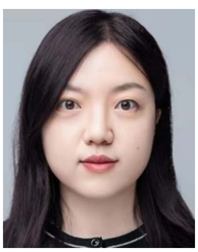
[38] Y. Jiang, Y. Yang, L. Xia, and C. Huang, "DiffKG: Knowledge graph diffusion model for recommendation," in *Proc. 17th ACM Int. Conf. Web Search Data Mining*, 2024, pp. 313–321.

[39] Z. Li, L. Xia, and C. Huang, "RecDiff: Diffusion model for social recommendation," in *Proc. 33rd ACM Int. Conf. Inf. Knowl. Manage.*, 2024, pp. 1346–1355.

[40] Z. Li, A. Sun, and C. Li, "DiffuRec: A diffusion model for sequential recommendation," *ACM Trans. Inf. Syst.*, vol. 42, no. 3, pp. 1–28, 2023.

[41] Z. Yang, J. Wu, Z. Wang, X. Wang, Y. Yuan, and X. He, "Generate what you prefer: Reshaping sequential recommendation via guided diffusion," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, vol. 36, pp. 24247–24261.

[42] H. Ma, Y. Yang, L. Meng, R. Xie, and X. Meng, "Multimodal conditioned diffusion model for recommendation," in *Proc. Companion Proc. ACM Web Conf. 2024*, 2024, pp. 1733–1740.
[43] Y. Jiang, L. Xia, W. Wei, D. Luo, K. Lin, and C. Huang, "DiffMM: Multi-modal diffusion model for recommendation," in *Proc. 32nd ACM Int. Conf. Multimedia*, 2024, pp. 7591–7599.
[44] J. Chen, H.-r. Fang, and Y. Saad, "Fast approximate kNN graph construction for high dimensional data via recursive Lanczos bisection," *J. Mach. Learn. Res.*, vol. 10, pp. 1989–2012, Dec. 2009.
[45] J. Ho and T. Salimans, "Classifier-free diffusion guidance," 2022, *arXiv:2207.12598*.
[46] S. Rendle, C. Freudenthaler, Z. Gantner, and L. Schmidt-Thieme, "BPR: Bayesian personalized ranking from implicit feedback," in *Proc. 25th Conf. Uncertainty Artif. Intell.*, 2009, pp. 452–461.
[47] Z. Guo, J. Li, G. Li, C. Wang, S. Shi, and B. Ruan, "LGMRec: Local and global graph learning for multimodal recommendation," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 8, pp. 8454–8462.
[48] X. Zhou, "MMRec: Simplifying multimodal recommendation," in *Proc. 5th ACM Int. Conf. Multimedia Asia Workshops*, 2023, pp. 1–2.
[49] L. v. d. Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, 2008.
[50] C. Wang et al., "Towards representation alignment and uniformity in collaborative filtering," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2022, pp. 1816–1825.



Jie Guo (Senior Member, IEEE) received the BS degree in communication engineering from Zhengzhou University, Zhengzhou, China, in 2011, and the PhD degree from Xidian University, Xi'an, China, in 2017. She is currently an associate professor with Xidian University. From 2015 to 2016, she got the State Scholarship Fund from China Scholarship Council to be an exchange Ph.D. Student with Carleton University, Ottawa, ON, Canada. Her research interests include information fusion, deep learning, and cross-modal retrieval.

Ziyuan Guo received the BE degree in electronic information engineering from Zhengzhou University, Zhengzhou, China, in 2023. He is currently working toward the MS degree in electronic information engineering with the School of Telecommunications Engineering, Xidian University, Xi'an, China. His research interests include deep learning and recommender systems.

Bin Song (Senior Member, IEEE) received the BS, MS, and PhD degrees in communication and information systems from Xidian University, Xi'an, China, in 1996, 1999, and 2002, respectively. He is currently a professor with Xidian University, Xi'an, China. He has authored more than 100 journal papers or conference papers and 50 patents. His research interests include multimedia communication, multimodal data fusion, content-based image recognition and machine learning, reinforcement learning, Internet of Things, Big Data, and recommender systems.

Peidong Zhang received the BE degree in electronic information engineering from Xidian University, Xi'an, China, in 2025. He is currently working toward the MS degree in electronic information engineering with the School of Telecommunications Engineering, Xidian University, Xi'an, China. His research interests include deep learning and recommender systems.